

SESSION 1

RAD-541 Magnetic Resonance Imaging II

The Main Magnetic Field, Protons and Larmor Frequency

Building the physical foundation for spatial encoding in MRI

Course	Magnetic Resonance Imaging II
Date	Sept, 2026
Lecturer	A. A. Zubair
Suggested study time	240–300 minutes

Learning Objectives

By the end of this session, you should be able to:

1. describe the purpose of the main magnetic field, B_0 , in MRI;
2. explain what is meant by a **homogeneous magnetic field**;
3. describe the behaviour of hydrogen protons before and after a patient enters B_0 ;
4. explain parallel and antiparallel alignment and how they produce net magnetization;
5. describe proton precession in simple physical terms;
6. state and apply the Larmor relationship between magnetic-field strength and precessional frequency; and
7. explain why these ideas are essential before learning slice selection and spatial encoding.

The central idea

MRI begins with a strong, highly uniform magnetic field called B_0 . Hydrogen protons placed in this field do not simply “point toward the magnet.” They adopt preferred energy states and **precess** around the direction of B_0 at a predictable frequency. That frequency is the **Larmor frequency**. Later, MRI spatial encoding works by deliberately changing magnetic field strength with gradients so that Larmor frequency changes with position.

1 Session Orientation

This session establishes the reference physics used throughout the spatial-encoding series. The sequence is deliberately simple:

$B_0 \longrightarrow$ energy states and $M_0 \longrightarrow$ precession $\longrightarrow$ Larmor frequency
 $\longrightarrow$ resonance $\longrightarrow$ spatial encoding.

The important question is not yet “how is the image reconstructed?” It is **why magnetic-field strength can be translated into frequency**. Once that relationship is clear, gradient fields and slice selection become much easier to understand.

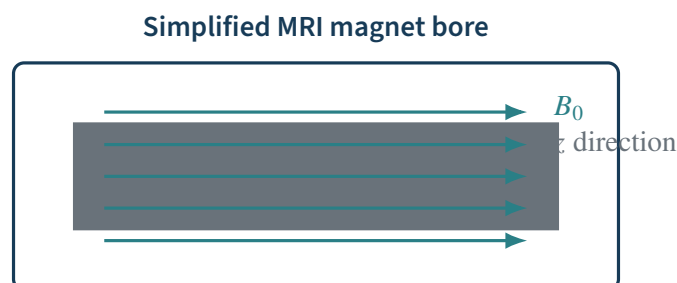
2 Why start with the main magnetic field?

MRI spatial encoding is often introduced using slice, phase and frequency gradients. However, gradients only make sense after one fact is understood: **proton frequency depends on magnetic-field strength**. The main magnetic field provides the reference field from which every later gradient effect is created.

A typical MRI scanner therefore does two important things before spatial encoding begins:

1. it establishes a strong static field, B_0 ; and
2. it tries to make that field as uniform as possible over the region being imaged.

For most clinical cylindrical MRI systems, B_0 is directed along the long axis of the scanner bore. By convention, that direction is called the **z-axis**. The scanner **isocentre** is the central reference region where the imaging system is designed and calibrated to provide its most controlled magnetic-field and gradient performance.

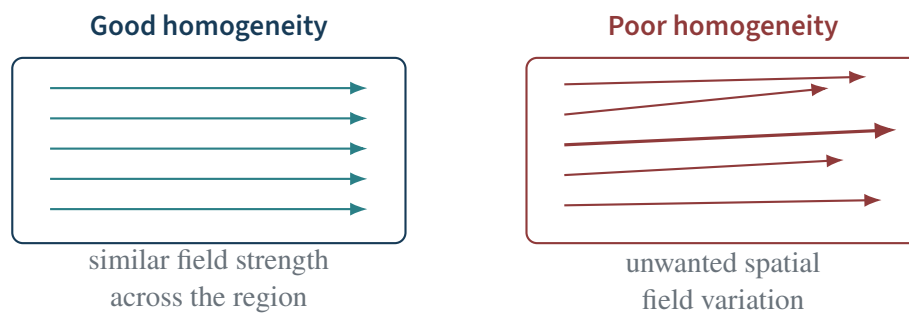


Ideally, field strength is the same throughout the imaging region.

3 What does “homogeneous magnetic field” mean?

A **homogeneous magnetic field** is one whose strength is nearly the same at every point within the region of interest. If two identical hydrogen protons are placed at different locations in a perfectly homogeneous B_0 , they experience essentially the same field and therefore have essentially the same Larmor frequency.

Homogeneity matters because MRI later uses very small, deliberately created frequency and phase differences to encode position. Unwanted field variation from the magnet, patient anatomy or magnetic susceptibility can create additional off-resonance frequencies that were *not* produced by the encoding gradients.



Radiography / MRI practice

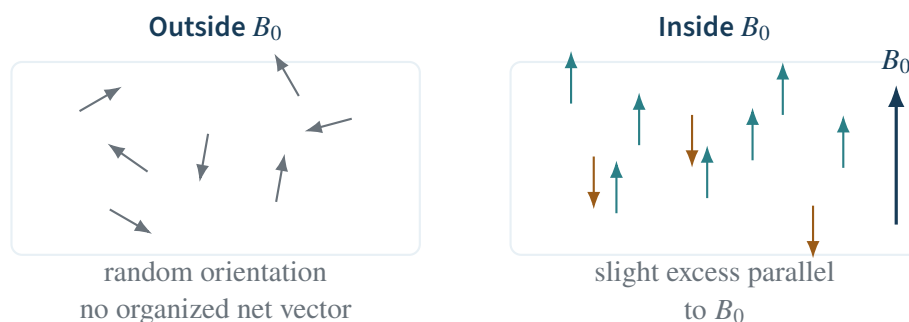
Field homogeneity affects image quality. Poor homogeneity can contribute to geometric distortion, signal loss, unreliable spectral fat suppression and other off-resonance artifacts. MRI systems therefore use **shimming**-adjustments that reduce unwanted field variation-to improve uniformity over the region being examined. The practical principle is: **B_0 should be as uniform as possible first; gradients then add controlled, temporary spatial variation for encoding.**

4 Why hydrogen protons are important in MRI

The human body contains abundant hydrogen, especially in water and fat. The nucleus of ordinary hydrogen contains a single proton and has a useful nuclear magnetic moment. This is one reason hydrogen is the principal nucleus used in routine clinical MRI.

The word **spin** describes an intrinsic quantum property of the proton; the proton should not be imagined literally as a tiny ball rotating in space. For introductory MRI physics, however, it is useful to represent its magnetic moment as a small compass-like vector that responds to an external magnetic field.

Outside the scanner there is no strong field imposing a preferred direction, so the magnetic moments are effectively randomly oriented. Their vector contributions cancel on average and there is no organized macroscopic magnetization available for MRI.



5 What happens when protons enter B_0 ?

Once tissue is placed in B_0 , proton magnetic moments can be described using two main energy states.

Parallel state

The proton magnetic moment is oriented generally with B_0 . This is the lower-energy state.

Antiparallel state

The proton magnetic moment is oriented generally against B_0 . This is the higher-energy state.

There are protons in both states, but thermal equilibrium produces a small excess in the lower-energy parallel state. The energy separation between the two states increases with magnetic-field strength, which helps explain why stronger fields can produce a larger equilibrium magnetization.

Although the population difference is small as a fraction of all protons, tissue contains an enormous number of hydrogen nuclei. Their combined longitudinal contribution therefore produces a measurable macroscopic vector called the **net magnetization vector**. By convention, the equilibrium net magnetization points along B_0 , or the z -axis, and is written as M_0 .

Do not overstate the alignment

It is incorrect to say that “all protons align with B_0 .” MRI contains large numbers of protons in both lower- and higher-energy states. The useful net magnetization exists because there is only a **small population excess** in the lower-energy state.

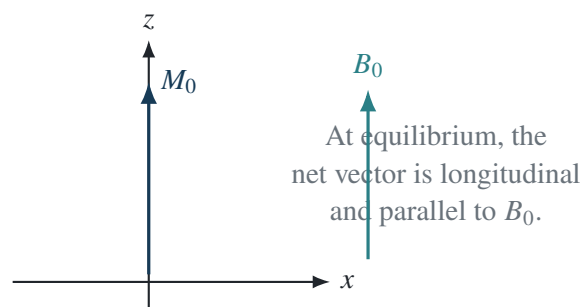
6 Net magnetization: from many protons to one useful vector

MRI does not track individual protons one by one. Instead, it describes the combined behaviour of a very large ensemble. When the longitudinal magnetic components are summed, the result is M_0 .

At equilibrium:

- the net longitudinal magnetization is non-zero;
- individual magnetic moments precess, but their transverse phases are not synchronized, so the transverse components largely cancel; and
- M_0 points along B_0 .

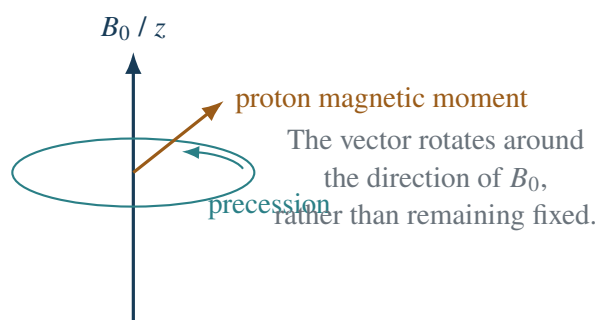
This distinction is important: **precession already exists before RF excitation, but coherent transverse magnetization does not**. RF excitation later creates an organized transverse component that can induce a measurable MR signal.



7 Protons do not stand still: precession

A proton magnetic moment in B_0 does not simply remain fixed along one direction. It undergoes a characteristic rotational motion around the axis of the main field called **precession**.

A useful analogy is a spinning top whose axis sweeps around the vertical direction. The analogy is not exact quantum mechanics, but it correctly emphasizes the feature MRI uses: the magnetic moment has a well-defined precessional frequency around B_0 .



The speed of this precession is not arbitrary. It is determined by the magnetic-field strength and the type of nucleus. For hydrogen, the relationship is given by the Larmor equation.

8 The Larmor frequency

The angular precessional frequency is

$$\omega_0 = \gamma B_0,$$

where ω_0 is angular frequency in radians per second, γ is the gyromagnetic ratio and B_0 is magnetic-field strength.

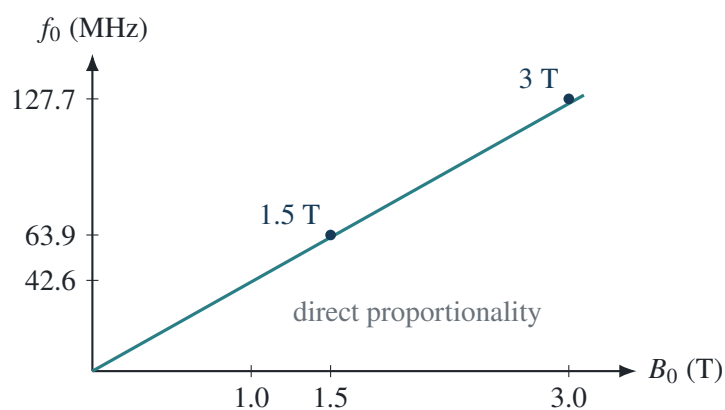
Clinical MRI discussions usually use ordinary frequency, f_0 , in hertz:

$$f_0 = \frac{\gamma}{2\pi} B_0.$$

For hydrogen,

$$\frac{\gamma}{2\pi} \approx 42.58 \text{ MHz/T}.$$

Thus, for a given nucleus, Larmor frequency is directly proportional to magnetic-field strength. The gyromagnetic ratio is nucleus-specific; the familiar 42.58 MHz/T value applies to hydrogen.



Larmor frequency at 1.5 T

For hydrogen in a 1.5 T scanner,

$$f_0 = 42.58 \times 1.5 \approx 63.87 \text{ MHz}.$$

So hydrogen protons near isocentre precess at approximately **63.9 MHz**.

Larmor frequency at 3 T

For hydrogen in a 3 T scanner,

$$f_0 = 42.58 \times 3 \approx 127.74 \text{ MHz.}$$

Thus, doubling field strength from 1.5 T to 3 T approximately doubles the hydrogen Larmor frequency.

Exam relationship

For the same nucleus,

$$B_0 \uparrow \implies f_0 \uparrow .$$

The relationship is linear. This fact becomes the basis of frequency-based spatial encoding when gradient fields are introduced.

9 Why field homogeneity and Larmor frequency are connected

Suppose the scanner intends every proton in a small region to experience exactly 1.5 T. If local susceptibility or imperfect shimming makes some positions experience a slightly higher or lower field, their Larmor frequencies also shift. The scanner then receives frequency differences that were not intentionally created by the spatial-encoding gradients. These unwanted shifts are called **off-resonance** effects.

This is why the MRI system needs two apparently opposite capabilities:

1. a very homogeneous B_0 as the stable reference field; and
2. controlled gradient fields that deliberately make magnetic field strength vary with position during encoding.

The bridge to Session 2

Spatial encoding is possible because a gradient makes B depend on position. Once B varies with position, Larmor frequency also varies with position. The scanner can then use frequency to help identify where signal came from. Session 2 develops this idea through gradient fields and slice selection.

10 Field strength in clinical MRI: what should the radiographer understand?

A radiographer does not need to calculate the magnet design, but should understand what changes when field strength changes.

Concept	1.5 T example	3 T example
Main field B_0	1.5 tesla	3 tesla
Hydrogen Larmor frequency	about 63.9 MHz	about 127.7 MHz
Frequency-field relationship	lower field, lower frequency	higher field, higher frequency
Practical message	RF system operates around a lower resonance frequency	RF system operates around a higher resonance frequency

Higher field strength also increases the equilibrium population difference and therefore tends to increase the available magnetization and signal potential, although many practical trade-offs accompany high-field MRI. For this session, retain the two foundations: **field strength determines resonance frequency, and a stronger B_0 also changes the equilibrium spin system.**

11 A first look at resonance

An RF pulse produces an oscillating magnetic field, commonly denoted B_1 , that is applied perpendicular to B_0 . When the RF frequency matches the hydrogen Larmor frequency, energy can be transferred efficiently to the spin system. This matching condition is called **resonance**.

For this series, remember the chain:

$$B_0 \rightarrow \text{Larmor frequency} \rightarrow \text{RF at matching frequency} \rightarrow \text{excitation}.$$

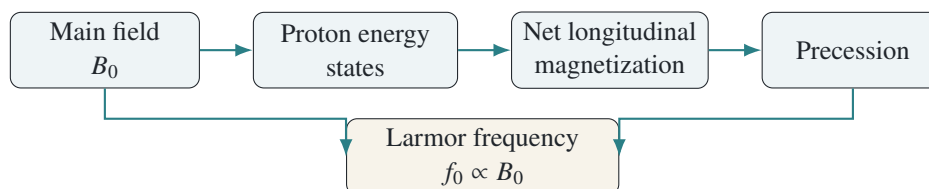
In Session 2, a slice-select gradient will deliberately make Larmor frequency depend on position. An RF pulse with a chosen centre frequency and bandwidth can then excite a selected range of positions, producing slice selection.

Common misconceptions

- **“All protons align with B_0 .”** No. There are populations in both energy states; a small excess produces net longitudinal magnetization.
- **“The proton physically orbits around B_0 .”** No. The useful model is precession of its magnetic moment around the field direction; nuclear spin itself is a quantum property.
- **“Larmor frequency is fixed for hydrogen.”** It is fixed only for a given magnetic-field strength. Change B , and the frequency changes.
- **“A perfectly uniform field would make spatial encoding impossible.”** The uniform field is the reference. Gradients temporarily add controlled spatial variation when encoding is required.

12 Putting the whole session together

The physical chain for Session 1 is:



Key Takeaways

- B_0 is the strong static reference field of the MRI scanner and is conventionally aligned with the z -axis.
- A homogeneous field has nearly the same strength throughout the imaging region.
- Hydrogen protons behave as tiny magnetic dipoles and occupy parallel and antiparallel energy states in B_0 .
- A small excess of lower-energy protons produces net longitudinal magnetization, M_0 .
- Proton magnetic moments precess around B_0 .
- Larmor frequency is directly proportional to magnetic-field strength.
- For hydrogen, $f_0 \approx 42.58 \text{ MHz/T} \times B_0$.
- Spatial encoding becomes possible when gradient fields deliberately make magnetic field strength, and therefore Larmor frequency, depend on position.

Key Terms

B_0

The main static magnetic field of the MRI scanner.

Homogeneity

Uniformity of magnetic-field strength within the imaging region.

Isocentre

Central scanner reference region where field and gradient performance are designed to be well controlled.

Off-resonance

Frequency shift caused by unwanted local magnetic-field variation rather than intentional encoding.

Hydrogen proton

The MRI-active nucleus most commonly used in clinical imaging.

Parallel state

Lower-energy orientation relative to B_0 .

Antiparallel state

Higher-energy orientation relative to B_0 .

Net magnetization, M_0

The combined magnetic vector produced by the slight excess of lower-energy protons.

Precession

Rotation of a magnetic moment around the direction of B_0 .

Gyromagnetic ratio, γ

Constant relating magnetic-field strength to precessional frequency for a particular nucleus.

Larmor frequency

Precessional frequency of a nucleus in a magnetic field.

Shimming

Adjustment used to improve magnetic-field homogeneity.

B_1

Oscillating RF magnetic field used to excite the spin system.

Resonance

Efficient RF interaction when RF frequency matches the Larmor frequency.

Self-Assessment Questions

1. What is the main function of B_0 in MRI?
2. Define magnetic-field homogeneity and state two reasons why it matters in MRI.
3. Explain why the “tiny spinning compass” description is useful but not literally correct.
4. Distinguish between the parallel and antiparallel energy states.

5. Why does a net magnetization vector exist if protons occupy both energy states?
6. What is proton precession, and why is there no coherent net transverse magnetization at equilibrium?
7. State the Larmor equation in terms of ordinary frequency and magnetic-field strength.
8. Calculate the approximate hydrogen Larmor frequency at 1.0 T, 1.5 T and 3.0 T.
9. Why can poor B_0 homogeneity create problems for spatial encoding?
10. In one sentence, explain how the Larmor relationship prepares the student for understanding gradient fields.

Series navigation

Next session: Gradient Fields, Imaging Axes and Slice Selection. We will use the Larmor relationship from this session to explain how a gradient creates position-dependent frequency and how the scanner selects a slice.

